

QUESTION 1 (20 marks)

- (a) Find the coordinates of point C(x,y) which divides the interval joining A (-1,3) and B(5,7) **externally** in the ratio 2:3. 3
- (b) Solve and graph on number line $\frac{x+1}{x-3} \geq 2$ 4
- (c) Find the acute angle between the line $y=2x-1$ and $2y+3x=4$ 4
- (d) Show that $\tan 75^\circ = 2 + \sqrt{3}$ 3
- (e) Differentiate $x^2\sqrt{x^2+1}$ 3
- (f) (i) Sketch the graph of $y=|x-2|$ and $y=-x$ on same set of axes 2
- (ii) For what values of x is $|x-2| \leq x$ 1

QUESTION 2 (22 marks)

- (a) A polynomial $P(x) = 9x^3 - 25x^2 + 10bx - b^2$
- (i) Find the remainder when P(x) is divided by x-1 2
- (ii) Find the value(s) of b if P(x) is divisible by x-1 3
- (b) If α, β and γ are the roots of $2x^3 + 3x^2 - 5x + 1 = 0$
Find
- (i) $\alpha + \beta + \gamma$ (ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ (iii) $\alpha\beta\gamma$ 3
- (iv) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
- (v) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (vi) $(\alpha\beta)^2\gamma + \alpha(\beta\gamma)^2 + (\alpha\gamma)^2\beta$ 2
- (c) (i) Factorise fully the polynomial $P(x) = x^3 - 6x^2 - 9x + 14$ 3
- (ii) Solve $P(x)=0$ 1
- (iii) Graph $y = P(x)$ 2
- (d) Simplify $\frac{5^n + 5^{n+2}}{5^{n+1}}$ 2

(i) $2\cos 2\theta + 1 = 0$ 4

$$(ii) 3 \tan^2 \theta = \sec^2 \theta + 2 \quad 4$$

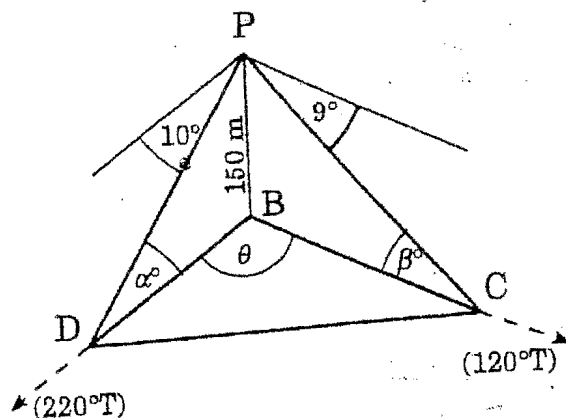
(iii) $\cot \theta + \tan \theta = 2 \operatorname{cosec} \theta$ 4

(b) If $t = \tan \frac{x}{2}$, prove that

$$\sin x = \frac{2t}{1+t^2} \quad 2$$

(c) Solve $5\sin x + 12\cos x = 2$ for $0 \leq \theta \leq 360^\circ$

(d) A ship is observed at the point C from the top of a cliff BP, 150m high in the direction $120^\circ T$ from B, at an angle of depression 9° , 10 minutes later, it is seen in the direction $220^\circ T$ from B at an angle of depression of 10° .



Let $\angle PDB = \alpha^\circ$, $\angle PCB = \beta^\circ$ and $\angle CBD = \theta^\circ$

(i) Write down the values of α, β and θ in degrees 3

(ii) Show that $BD = 150 \tan 80^\circ$, and $BC = 150 \tan 81^\circ$ 3

(iii) Hence show that $CD^2 = 150^2(\tan^2 80^\circ + \tan^2 81^\circ - 2 \tan 80^\circ \tan 81^\circ \cos 100^\circ)$ 3

(iv) Find CD 2

QUESTION 4 (19 marks)

- (a) A combination lock has 4 dials each with 10 digits. How many possible combinations are there? 1
- (b) How many 3 letter words can be made from NORWAY? 1
- (c) How many ways can the letters of BACHARACH be arranged using all the letters? 3
- (d) A team of 10 boys and 10 girls prefects is to be chosen from a group of 21 boys and 19 girls. Leave answers in unsimplified form.
 (i) How many different ways can the prefects be chosen? 3
- (e) If Katrina and Jonathan are hoping to be chosen, find the probability that
 (α) Both will be chosen? 2
 (β) Only one will be chosen? 3
- (f) To win Lotto you choose 6 numbers out of the numbers 1 to 45 different numbers
 (i) In how many ways can this be done 2
 (ii) Tas does not select the numbers 11,12,13,14,15 and 16 for Lotto as he says this combination would never win. Is he correct? Give brief reasons. 2
- (g) Find the number of ways in which 3 boys and 3 girls can be seated around a circular table so that no two people of the same sex sit next to each other. 2

QUESTION 5 (21 marks)

- (a) (i) Write down the coordinates of the focus, S, and the equation of the directrix of the parabola $x^2 = 4ay$ 2
 (ii) Find the equation of the tangent to $x^2 = 4ay$ at the point $P(2ap, ap^2)$ 3
 (iii) The tangent at P meets the directrix at Q. Find the coordinates of Q. 2
 (iv) Find the gradients of the lines PS and QS 2
 (v) Prove that PQ subtends a right angle at the focus. 2
- (b) (i) Show that the point A $(6p, 3p^2)$ lies on the parabola $x^2 = 12y$ 2
 (ii) If B is the point $(6q, 3q^2)$. Find the equation of AB 3
 (iii) If AB passes through the point C(8,0) show that $4(p+q) = 3pq$ 2
 (iv) Write down the coordinates of the midpoint M of AB 2
 (v) Find the locus of M 3

Yr 11 Yearly Examination. Ext 1

QUESTION 1

(a) $(-1, 3)$ $(5, 7)$

$2 \div -3$

1 for -ve

$\frac{3+10}{-1}, \frac{-9+14}{-1}$

1 for each point.

$(-13, -5)$

If answer $(\frac{7}{5}, \frac{23}{5})$ 2 marks

(b) $x \neq 3$ ①

$x+1 = 2x-6$

$7 = x$ ①



$x=0$ $-\frac{1}{3} \geq 2 \times$

$x=4$ $5 \geq 2 \checkmark$

$x=10$ $\frac{11}{7} \geq 2 \times$

$3 < x \leq 7$ ①

(c) $m_1 = 2$ $2y = 4 - 3x$

① ②

$y = 2 - \frac{3}{2}x$

$m_2 = -\frac{3}{2}$ ①

$\tan \theta = \left| \frac{-\frac{3}{2} - 2}{1 + \frac{-3}{2} \times 2} \right|$ ①

$\tan \theta = 1.75$

$\theta = 60^\circ 15'$ ①

Q2d) $\frac{5^n + 5^{n+2}}{5^{n+1}} = \frac{5^n(1+5^2)}{5^{n+1}}$
 $= \frac{26}{5}$

(d) $\tan(45^\circ + 30^\circ) = \frac{\tan 45 + \tan 30}{1 - \tan 45 \tan 30}$ ①

$= 1 + \frac{1}{\sqrt{3}}$
 $\frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$ ①

$= \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$

$= \frac{3+2\sqrt{3}+1}{2}$

$= \frac{4+2\sqrt{3}}{2}$ ①

$= 2 + \sqrt{3}$

(e) (i) $\left| \frac{m - \frac{1}{\sqrt{3}}}{1 + \frac{m}{\sqrt{3}}} \right| = \tan 75^\circ$

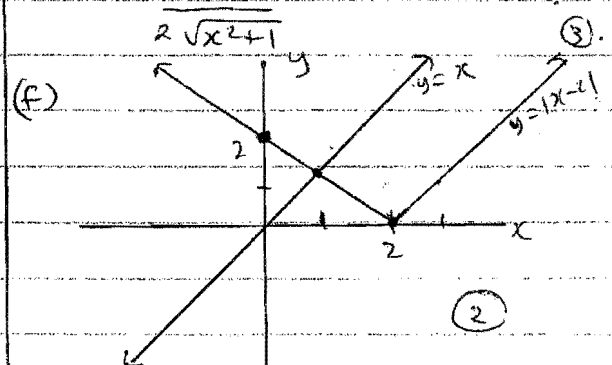
$\sqrt{3}m - 1 = (2 + \sqrt{3})(\sqrt{3} + m)$

$\sqrt{3}m - 1 = 2\sqrt{3} + 2m + 3 + \sqrt{3}m$

$\therefore 2m =$

(e) $\frac{dy}{dx} = x^2 \times \frac{1}{2}(x^2+1)^{-\frac{1}{2}} \times 2x + 2x\sqrt{x^2+1}$

$= \frac{2x^3}{2\sqrt{x^2+1}} + 2x\sqrt{x^2+1}$



(u) $x \geq 1$ ①

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QUESTION 3

(a) (i) $\cos 2\theta = -\frac{1}{2}$ ①

$2\theta = 120, 240, 480, 600$ ①

$\theta = 60, 120, 240, 300$ ①

2 marks if $\theta = 60, 120$ given.

(ii) $3\tan^2\theta = 1 + \tan^2\theta + 2$

$2\tan^2\theta = 3$

$\tan^2\theta = \frac{3}{2}$

$\tan\theta = \pm\sqrt{\frac{3}{2}}$

$\theta = 50^\circ 46', 230^\circ 46', 129^\circ 14'$

$309^\circ 14'$

(iii) $\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta} = \frac{2}{\sin\theta}$ ①

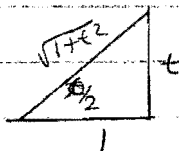
$\frac{\cos^2\theta + \sin^2\theta}{\sin\theta\cos\theta} = \frac{2}{\sin\theta}$ ①

$1 = 2\cos\theta$

$\cos\theta = \frac{1}{2}$ ①

$\theta = 60^\circ, 300^\circ$ ①

(b)



$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$

$= 2 \times \frac{t}{\sqrt{1+t^2}} \times \frac{1}{\sqrt{1+t^2}}$

$= 2 \times \frac{t}{\sqrt{1+t^2}} \times \frac{1}{\sqrt{1+t^2}}$

$\sin x = \frac{2t}{1+t^2}$ ②

(c) $5 \times \frac{2t}{1+t^2} + 12 \times \frac{1-t^2}{1+t^2} = 2$ ③

$10t + 12 - 12t^2 = 2 + 2t^2$

$14t^2 - 10t - 10 = 0$

$7t^2 - 5t - 5 = 0$

$t = \frac{5 \pm \sqrt{25+140}}{14}$

$\therefore \tan \frac{\theta}{2} = 1.275 \text{ or } -0.560$

$\frac{\theta}{2} = 51^\circ 53', 150^\circ 44'$

$\theta = 103^\circ 46', 301^\circ 28'$ ④

(d) (i) $\alpha = 10^\circ$ $\beta = 9^\circ$ $\theta = 100^\circ$ ③

(ii) $\frac{BD}{PB} = \tan(90-\alpha)$

$BD = 150 \tan 80$ similarly

$BC = 150 \tan(90-\alpha)$
 $= 150 \tan 81$ ③

(iii) using cosine rule

$CD^2 = (150 \tan 80)^2 + (150 \tan 81)^2 - 150^2 \tan 80 \tan 81 \cos 91$

$CD^2 = 150^2 (\tan^2 80 + \tan^2 81 - \tan 80 \tan 81 \cos 91)$ ③

(iv) $CD^2 = 1379$

②

Question 4

(a) $10 \times 10 \times 10 \times 10$

$= 10000$ (1)

(b) $\therefore$ (1)

$6 \times 5 \times 4 = 120$

(c) $\frac{9!}{3!2!2!}$

1 for 9!

2 if miss out factorial on bottom.

(d) ${}^{21}_9 C \times {}^{19}_{10} C =$ (2)

(ii) ${}^{20}_9 C \times {}^{18}_9 C =$ (2)

(iii) ${}^{20}_9 C \times {}^{18}_{10} C + {}^{20}_{10} C \times {}^{18}_9 C$ (3)

(e) (i) ${}^{45}_6 C = 8,445,060$ (2)

(ii) No since this selection has as good a chance as any. Since selection of numbers

is independent. (2)

(f) B'

G G

B B

G

$\therefore 3! \times 2! = 12$

Question 5

(a) (i) $(0, a)$ $y = -a$ (2)

(ii) $y = \frac{x^2}{4a}$

$\frac{dy}{dx} = \frac{2x}{4a}$

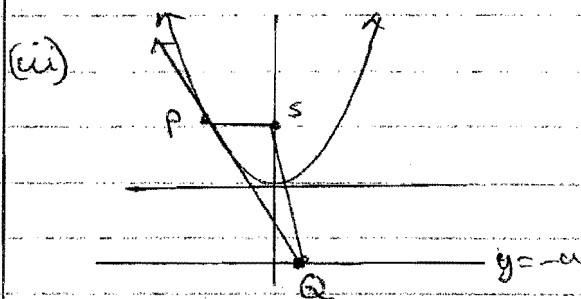
$= \frac{x}{2a}$

When $x = 2ap$

$\frac{dy}{dx} = p$

$y - ap^2 = p(x - 2ap)$ (3)

$y = px - ap^2$



When $y = -a$

$-a = px - ap^2$

$px = -a + ap^2$

$x = \frac{a(-1 + p^2)}{p}$

$\therefore \left(\frac{a(p^2 - 1)}{p}, -a \right)$ (2)

(iv) $S(0, a)$ $P(2ap, ap^2)$

$M_{PS} = \frac{ap^2 - a}{2ap}$

$= \frac{p^2 - 1}{2a}$

$= \frac{p^2 - 1}{2a}$

QUESTION 5.

$$M_{QS} = \frac{-a - a}{a(p^2 - 1)}$$

$$= \frac{-2a}{a(p^2 - 1)}$$

$$= \frac{-2p}{p^2 - 1} \quad (2)$$

$$(v) \therefore m_{PS} \times m_{QS}$$

$$= \frac{p^2 - 1}{2p} \times \frac{-2p}{p^2 - 1}$$

$$= -1$$

$$\therefore PS \perp QS \quad (2)$$

a right angled is subtended at focus.

$$y - 3q^2 = \left(\frac{p+q}{2}\right)(x - 6q)$$

$$2y - 6q^2 = px + qx - 6pq - 6q^2$$

$$\therefore (p+q)x - 2y - 6pq = 0 \quad (3)$$

$$(ii) \text{ when } x=8 \quad y=0$$

$$8(p+q) = 6pq \quad (2)$$

$$\therefore 4(p+q) = 3pq$$

$$(iv) M\left(\frac{6p+6q}{2}, \frac{3p^2+3q^2}{2}\right)$$

$$M\left(\frac{3p+3q}{2}, \frac{3p^2+3q^2}{2}\right)$$

(2)

$$(v) x = 3(p+q)$$

$$y = \frac{3}{2}(p^2+q^2)$$

$$(p+q)^2 = p^2 + q^2 + 2pq$$

$$\text{but } pq = \frac{4}{3}(p+q)$$

$$\therefore p^2 + q^2 = (p+q)^2 - \frac{8}{3}(p+q)$$

$$\therefore \frac{2}{3}y = p^2 + q^2$$

$$\frac{x}{3} = p+q$$

$$\therefore \frac{2}{3}y = \frac{x^2}{9} - \frac{8}{9}x$$

$$\therefore 6y = x^2 - 8x \quad (3)$$

$$\therefore 6y = x^2 - 8x \quad (3)$$

$$(b) x = 6p \quad y = 3p^2$$

$$LHS = 36p^2$$

$$RHS = 12 \times 3p^2 = 36p^2$$

(2)

$$\therefore (6p, 3p^2) \text{ is a point on } x^2 = 12y$$

$$(ii) m_{AB} = \frac{3q^2 - 3p^2}{6q - 6p} = \frac{3(q-p)(q+p)}{6(q-p)} = \frac{p+q}{2}$$

$$= \frac{p+q}{2}$$